

This is a very well written and worked out paper. Though the equations and formalism are well known and only *applied* to the pain bearing fiber case, the application is done well. The figures are high quality and illustrate well the mechanisms being considered. Limitations are acknowledged thoroughly and future work is outlined in several directions. It can have a major bearing on research in ephaptic coupling. We encourage the author to study higher dimension (two and three) formulations as well, which are also available in the literature (see Chawla-Morgera-Snider 2019 IEEE/ACM Transactions on Computational Biology and Bioinformatics and Chawla-Morgera Communications in Mathematical Biology and Neuroscience 2024). As a minor point which should be addressed before acceptance: please cite the 2003 work of Reutskiy-Rossoni-Tirozzi on conduction in bundles of demyelinated nerve fibers.